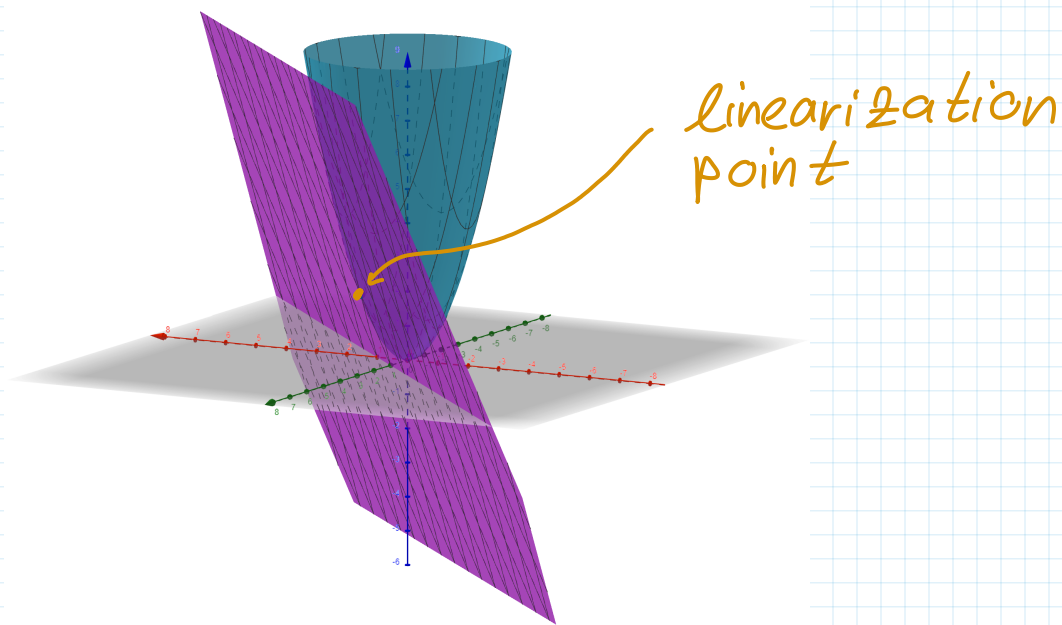


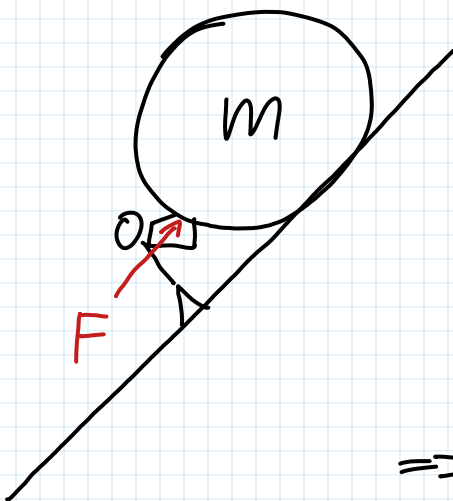
Where we left off...

$\Rightarrow$ Real world systems are almost always nonlinear.

$\Rightarrow$ In a certain "operational neighborhood" they can be approximated by linearization



• But some systems contain nonlinearities that are hard to neglect...

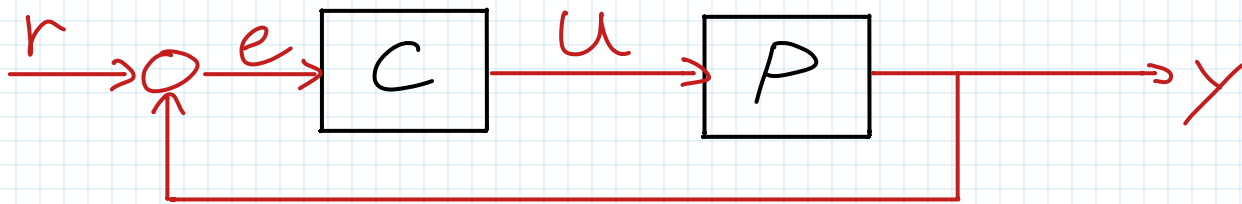


In theory, Sisyphos can always stabilize the boulder but in practice he can enact only limited forces!

$\Rightarrow$

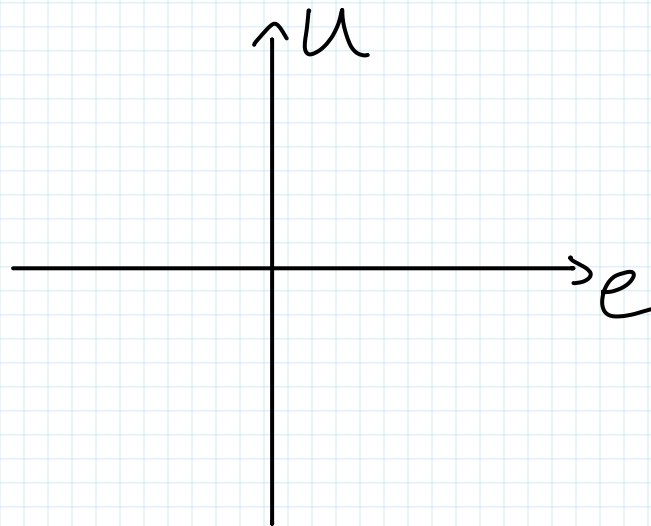
This is called a nonlinear nonlinearity.

Here: $u = F$
 $y = a$

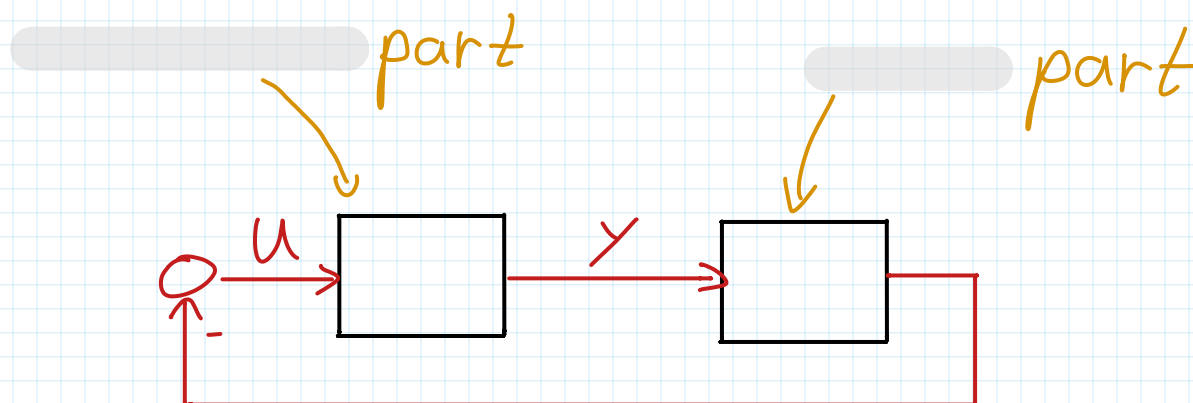


$\Rightarrow$ The error can grow arbitrarily large, but the input u can't!

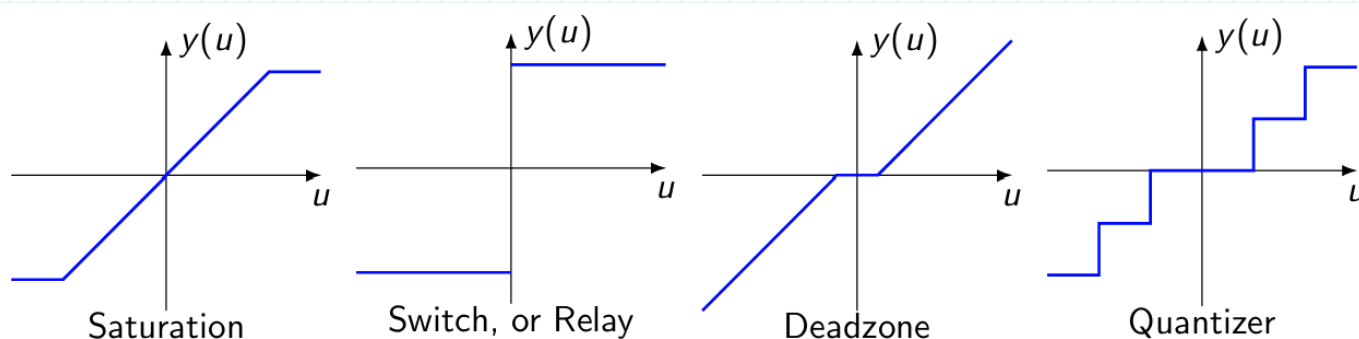
Hence Sysiphus' strength limitations must be modeled as such:



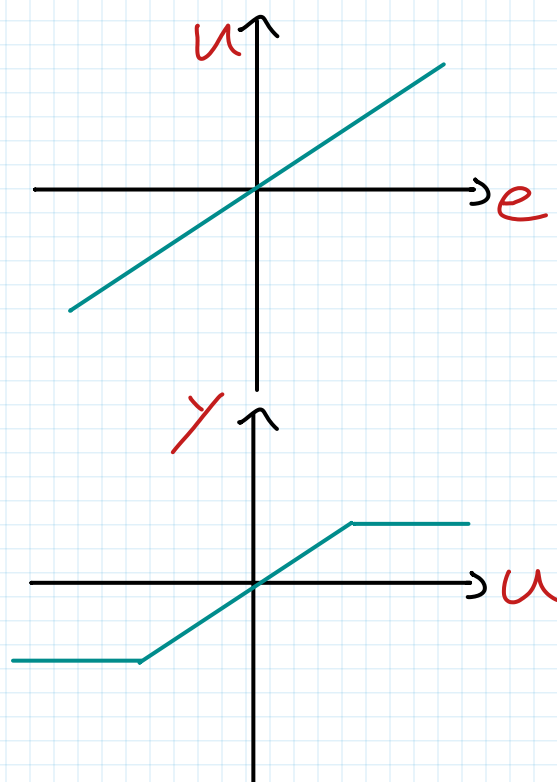
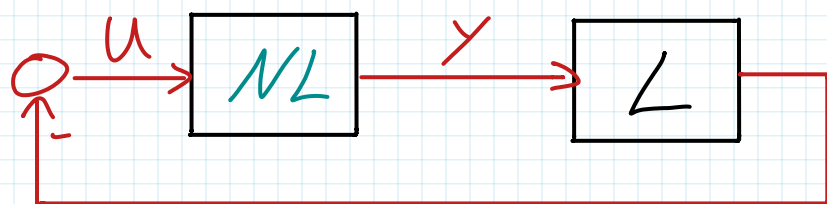
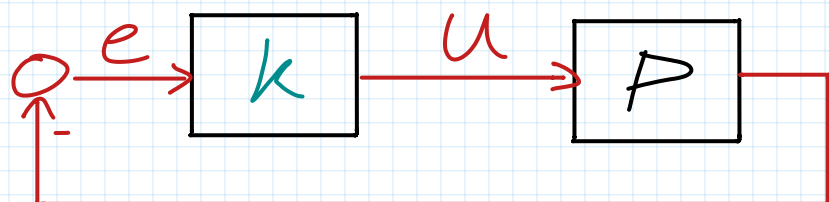
The trick will now be to model this nonlinear element separately and keep the linearized plant.



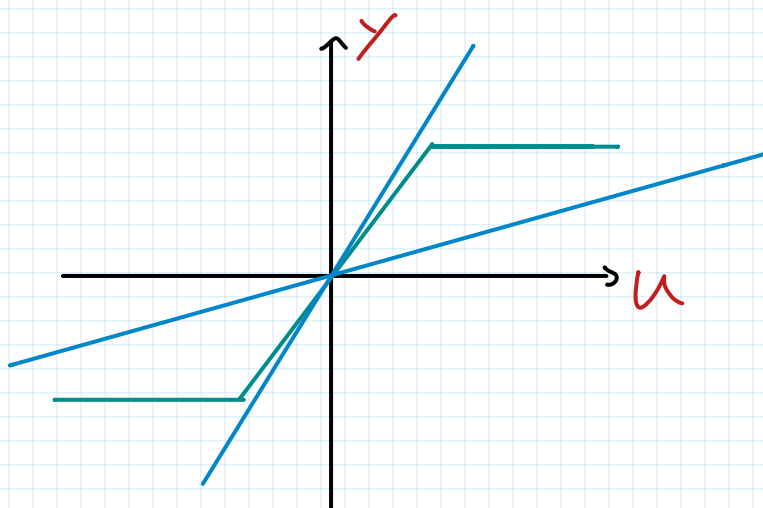
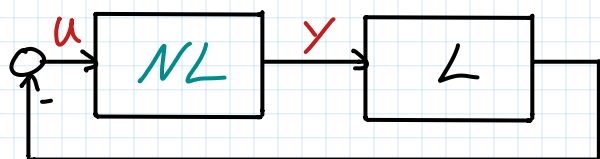
$\Rightarrow$ Common NL blocks include:



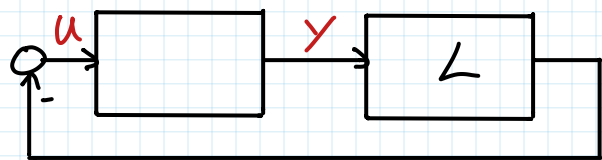
• The NL block acts like a nonlinear gain.



- Using this similarity to gains, we can assess stability of the system with the nonlinearity.
- A feedback system with a nonlinear element is if the closed loop is stable for all k that contain the possible .



- So now, if the loop is stable for any $k \in [k_1, k_2]$,

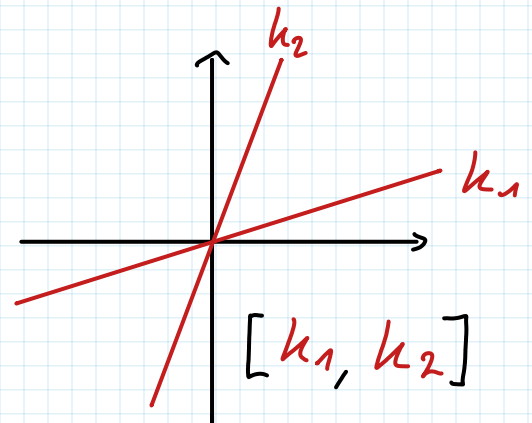


$\Rightarrow$ Its also stable with !

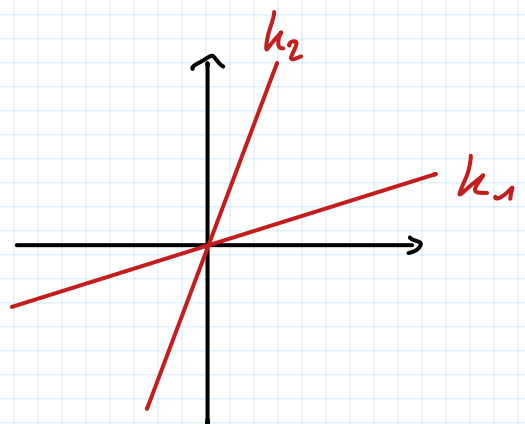
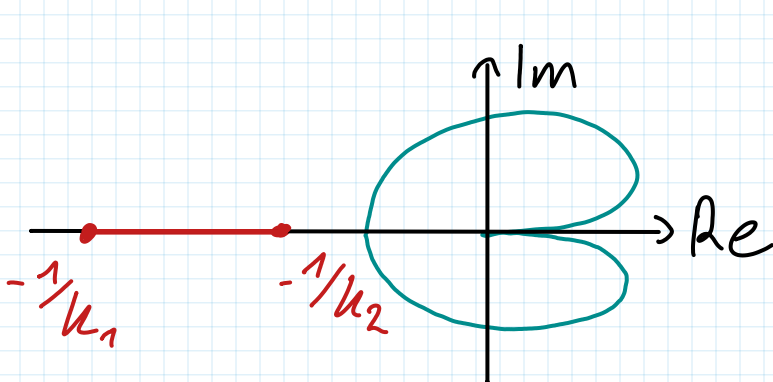
- Mathematically we need stability for:

$\Rightarrow$ What does this condition look like on a Nyquist-plot?

A hand-drawn coordinate system on blue grid paper. The horizontal axis is labeled 'Re' with an arrow pointing to the right. The vertical axis is labeled 'Im' with an arrow pointing upwards.



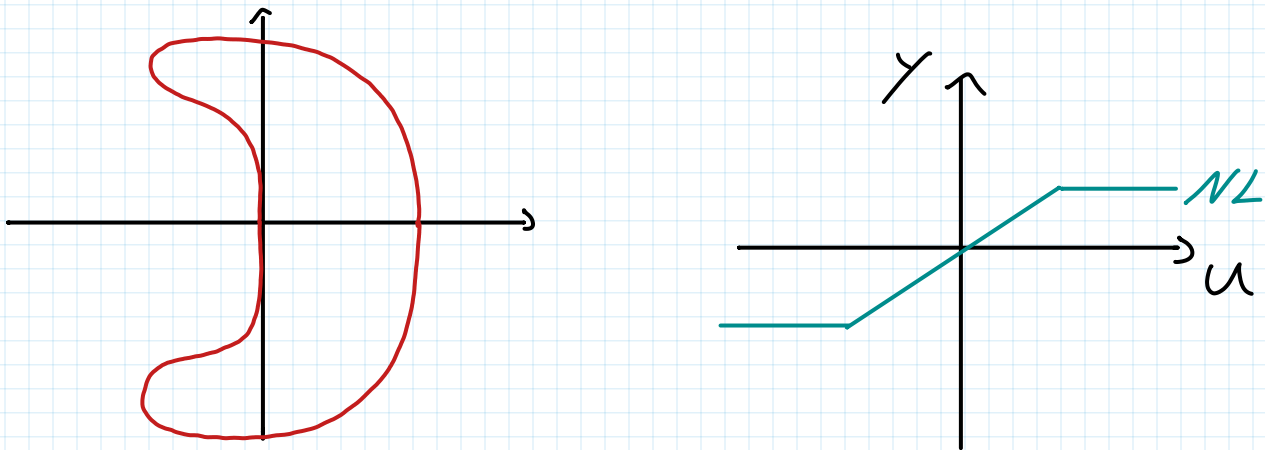
- The so called **boundary condition** is a **condition**:



$\Rightarrow$ For guaranteed stability, we must count encirclements of the circle.

Example:

• Assume an open-loop stable system with a saturation nonlinearity. The Nyquist plot is:



$\Rightarrow$ The Nyquist plot touches the circle!

• Necessary condition

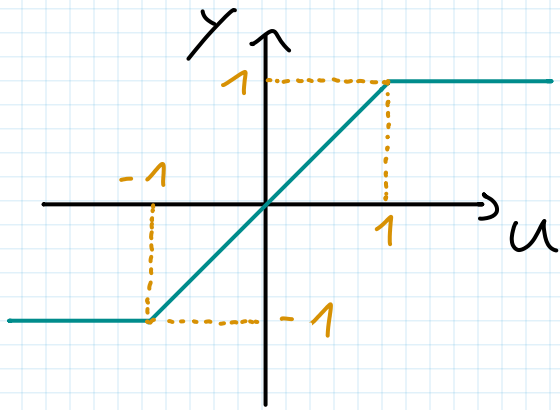
• Sufficient condition

$\Rightarrow$ We can't conclude if the closed-loop is stable!

We need a more precise way to describe the nonlinearity...

Describing functions

- A saturation nonlinearity is written as:



$$y = \begin{cases} 1 & u > 1 \\ u & -1 \leq u \leq 1 \\ -1 & u < -1 \end{cases}$$

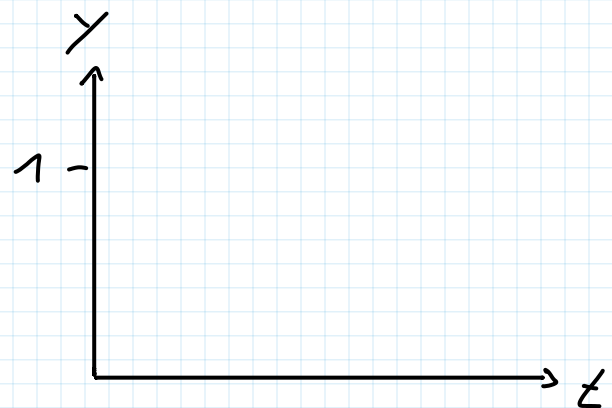
- In general our inputs u are sinusoids:

$$u(t) =$$

- The effect of the saturation is dependent on the input amplitude A .



$\Rightarrow$
Sat

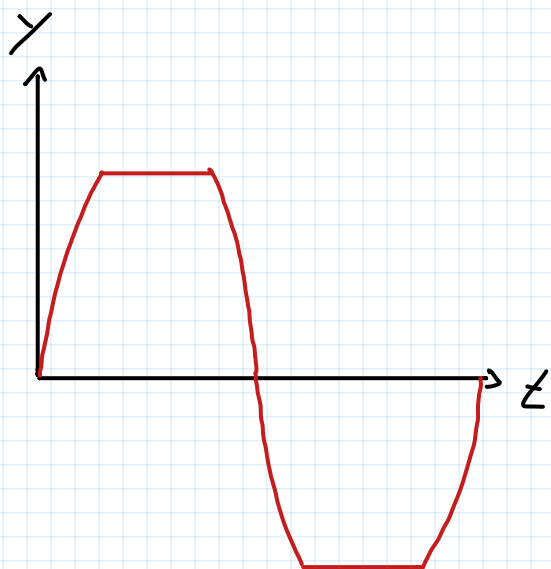


So the nonlinearity function

$$y(t) =$$

is also if $u(t)$ is periodic.

What could we use to describe a function explicitly that has such a weird shape??



Periodic functions like this can be constructed by of sinusoids.

$\Rightarrow$

We find the fourier representation with:

$$y(t) = \frac{a_0}{2} + \sum_{i=1}^{\infty} (a_i \cos(i\omega t) + b_i \sin(i\omega t))$$

$$a_i = \frac{1}{\pi} \int_{-\pi}^{\pi} y(t) \cos(i\omega t) dt$$

$$b_i = \frac{1}{\pi} \int_{-\pi}^{\pi} y(t) \sin(i\omega t) dt$$

• For the saturation case, the NL is
 so all terms

• In addition, we will approximate the NL-function with just the coefficient
 . So the NL is described as:

$$y(t) \approx$$

=> This simplification works, because real systems almost always reject signals anyway.

So we find the coefficient b_1 as:

$$b_1 =$$

$$\text{where } y(t) =$$

• clearly b_1 is dependent on

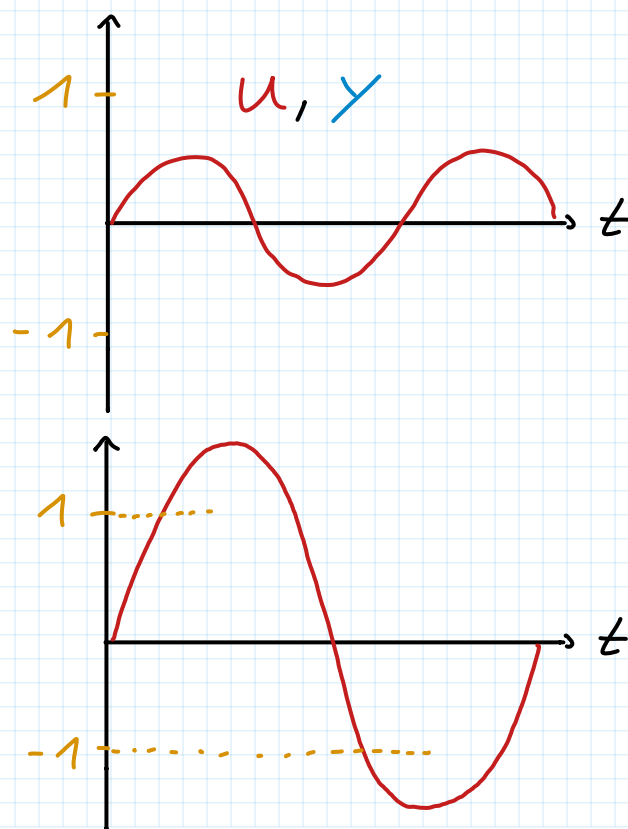
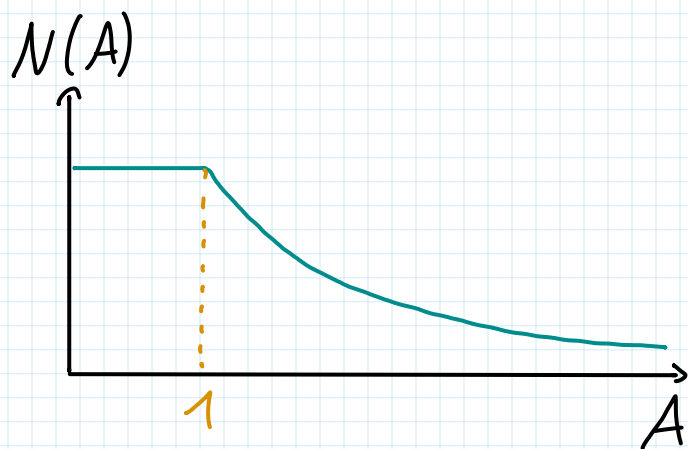
$$b_1(A) =$$

If we "normalize" $b_1(A)$ with A , we get the describing function of the saturation

$$N(A) =$$

If we solve this integral, we get different solutions depending on whether saturation occurs or not.

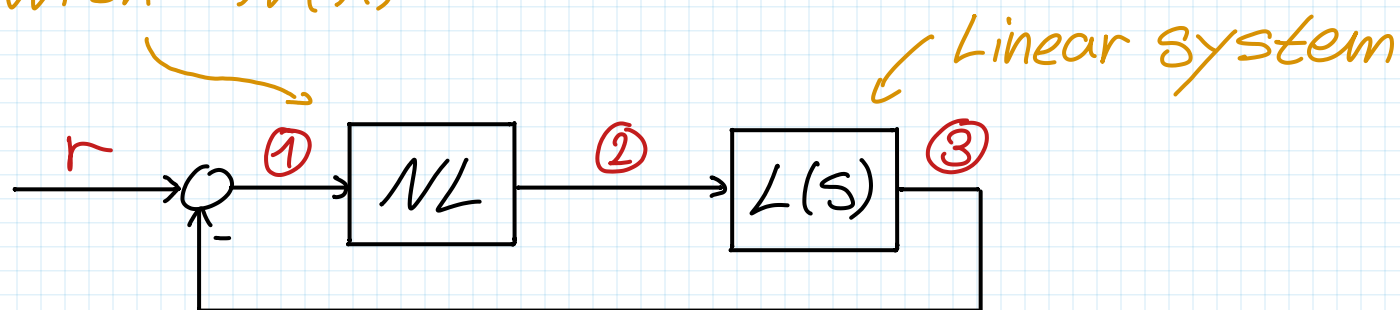
$$N(A) = \left\{ \begin{array}{l} \end{array} \right.$$



$\Rightarrow$ Here, $N(A)$ is a that
with .

Now we have a new method to describe nonlinearities. Can we now analyze stability using the describing functions?

Nonlinear system described
with: $N(A)$



• We can now approximate the open-loop as

Let's analyze how a signal propagates through this loop.

$\Rightarrow$ ① =

$\Rightarrow$ ② =

$\Rightarrow$ ③ =

If we now set , does A

• Increase?

• Decrease?

• Not change?

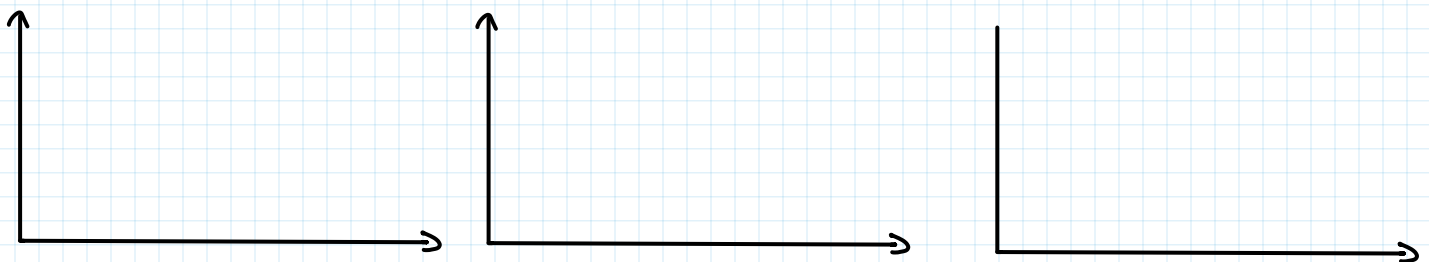
Since we assumed that ϕ has constant amplitude, our system has a self-sustaining oscillatory behaviour if

solving for the linear $L(s)$ we get:

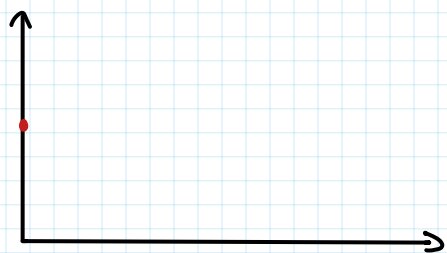
=> If this equation is fulfilled, a limit cycle is present.

This was very technical. Let's try and get an intuition for what a limit cycle is.

• With no reference inputs, our linear systems had to possible responses:

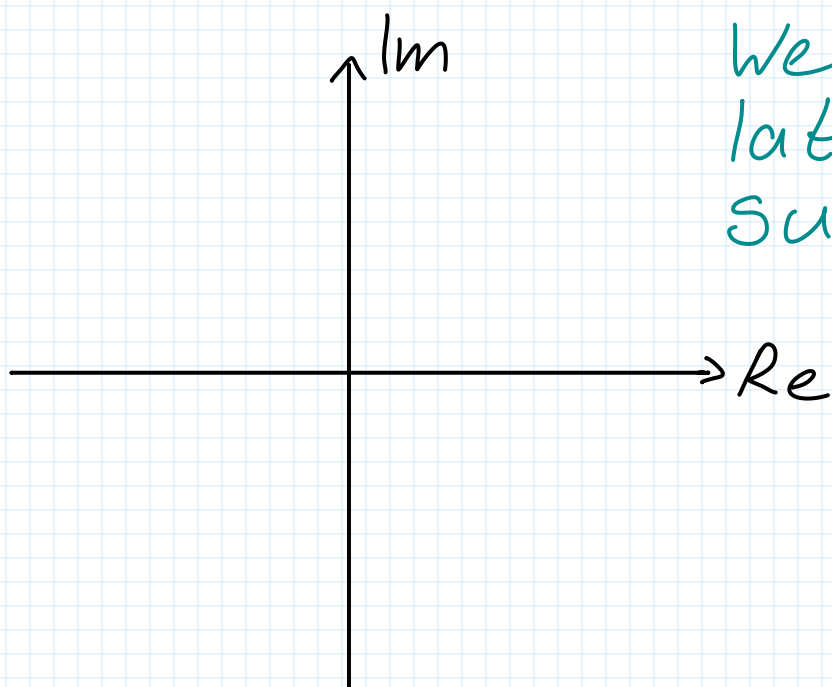


With nonlinear systems we can now also have stable, oscillations:



Remember, there is no reference input! The system is doing this on its own!

• So how do we check for limit cycles?
 => Check polar-plot / Nyquist:



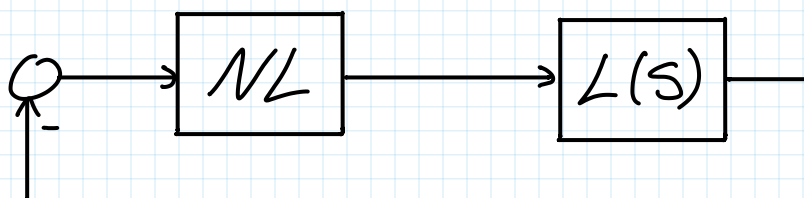
We said the oscillation is self-sustaining if:

=> Limit cycle at the

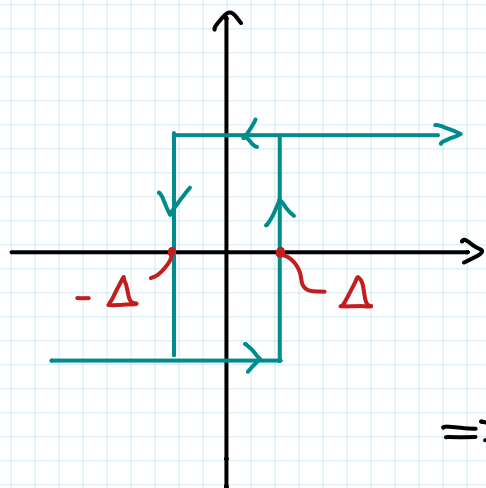
Example:

Schmitt-trigger

$$L(s) = \frac{1}{s(s+1)}$$



• Schmitt-trigger:

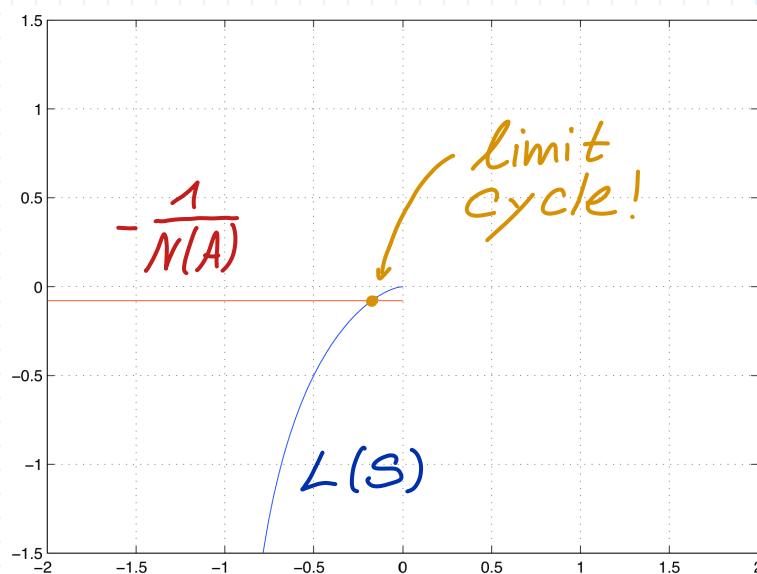


$$N(A) = \frac{4}{\pi A} \left(\sqrt{1 - \left(\frac{\Delta}{A} \right)^2} - j \frac{\Delta}{A} \right)$$

Δ : trigger value

$$\Rightarrow -\frac{1}{N(A)} = \frac{\pi}{4} \left(\sqrt{A^2 - \Delta^2} + j\Delta \right)$$

Drawing the polar plot:



• To find the ω and A that characterize the limit cycle we solve:

$$L(s) = -\frac{1}{N(A)} = \frac{\pi}{4} (\sqrt{A^2 - \Delta^2} + j\Delta)$$

• $L(j\omega) =$

Setting real and imaginary parts equal:

• Δ is given as:

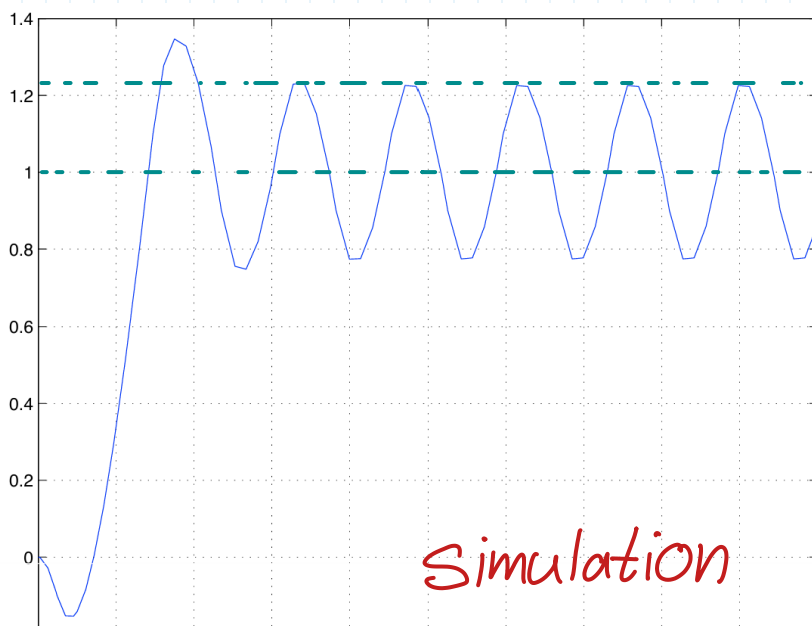
$$\Delta = 0, 1$$

We get:

$$\omega = 2,2 \frac{\text{rad}}{\text{s}}$$

$$A = 0,24$$

A

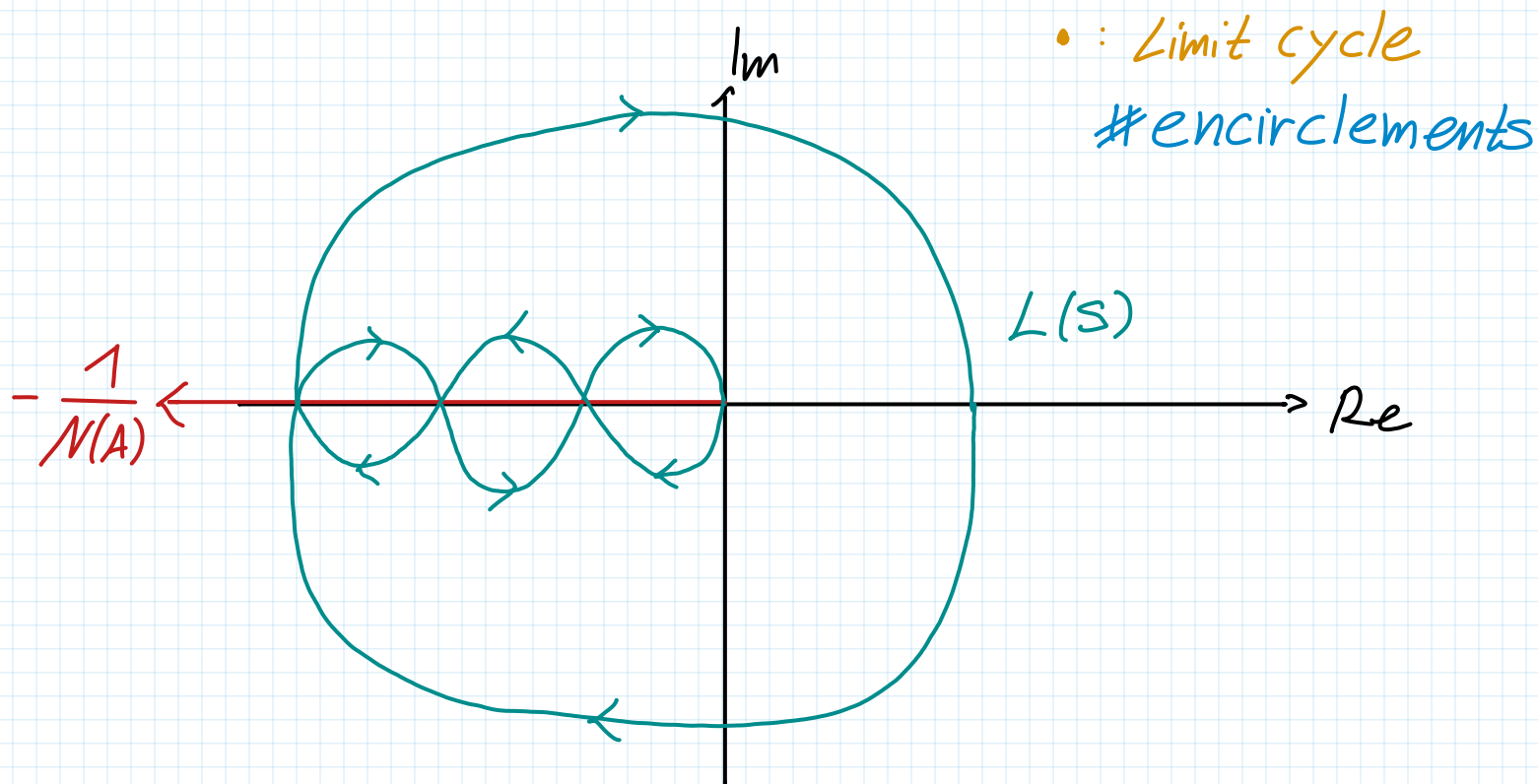


• More general: Limit cycles are stable if a small amplitude $\rightarrow$ causes a amplitude $\rightarrow$ and vice-versa.

$\Rightarrow$ We can also use the Nyquist plot to see if this is the case!

Example:

Given an open-loop stable system with Nyquist plot:



Let's look at an old exam question

Problem: Consider the closed-loop system T shown in Figure 23, where L is a linear time-invariant system and where NL is a saturation non-linearity.

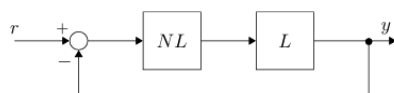


Figure 23: Standard feedback architecture. System L and saturation non-linearity denoted by NL .

The describing function $N(A)$ of the saturation non-linearity is given by,

$$N(A) = \frac{2}{\pi} \left[\arcsin\left(\frac{1}{A}\right) + \frac{1}{A} \sqrt{1 - \left(\frac{1}{A}\right)^2} \right].$$

It is known that $L(s)$ has a pole at the origin, i.e. at $s = 0$, and that $L(s)$ has one unstable pole. Figure 24 shows the Nyquist plot of $L(s)$ (solid line) together with $\frac{-1}{N(A)}$ (dashed line). Further, assume that all assumptions required for a describing function analysis are met.

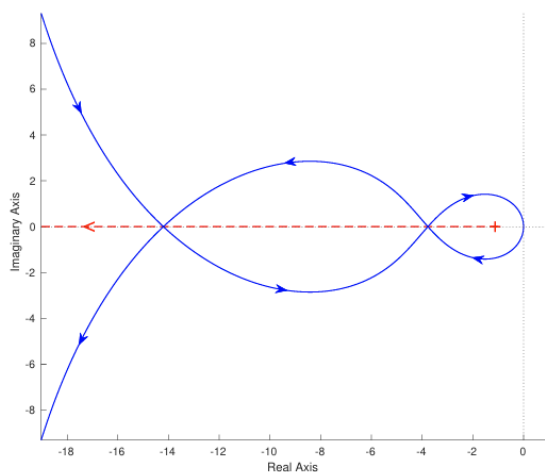


Figure 24: [Solid line] Nyquist plot of $L(s)$. [Dashed Line] Plot of $\frac{-1}{N(A)}$, where the arrow indicates the direction of $\frac{-1}{N(A)}$ as A increases.

Q40 (0.5 Points) Mark the correct answer.

First, without doing any stability analysis, what is the maximum number M of potential limit cycles that the closed-loop system T could support?

- ☐ A $M = 0$
☐ B $M = 2$

- ☐ C $M \rightarrow \infty$
☐ D $M = 1$

Q41 (1 Points) Mark the correct answer.

What is the number M_s of stable limit cycles that exist in the closed-loop system T ?

- ☐ A $M_s = 0$
☐ B $M_s = 2$

- ☐ C $M_s \rightarrow \infty$
☐ D $M_s = 1$

Q42 (1 Points) Mark all correct statements.

- ☐ A The circle criterion is a necessary and sufficient condition for absolute stability.
☐ B Describing functions can be seen as approximate transfer functions that represent the transfer function from sinusoidal inputs of amplitude A and frequency ω to the first harmonic of the output of the corresponding non-linearity.
☐ C Limit cycles are typically observed in linear systems.
☐ D The describing function of a static non-linearity is an amplitude dependent gain $N(A)$.

Finally, some quiz-questions!

Q1) Describing functions also work well in systems with multiple nonlinearities

☐ TRUE ☐ FALSE

Q2) Describing functions are a good fit for systems that have high-pass-filter behavior.

☐ TRUE ☐ FALSE

Q3) Using describing functions makes the total open-loop function L' dependent on ω and A .

☐ TRUE ☐ FALSE

Have a nice weekend ü